



Module-1: Introduction and Basic Definitions

Thermodynamics (NMEC103) : (Winter 2024-25)

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Important Thermodynamic Properties

p → *Pressure*

T → *Temperature*

V → *Volume*, ***v*** → *Specific Volume* = $\frac{V}{m}$

U → *Internal Energy*, ***u*** → *Specific internal energy* = $\frac{U}{m}$

H ≡ ***U + PV*** → *Enthalpy*, ***h*** ≡ ***u + pv*** → *Specific enthalpy* = $\frac{H}{m}$

S → *Entropy*, ***s*** → *Specific entropy* = $\frac{S}{m}$

G ≡ ***H - TS*** → *Gibb's Free Energy*, ***g*** ≡ ***h - Ts*** → *Specific G* = $\frac{G}{m}$

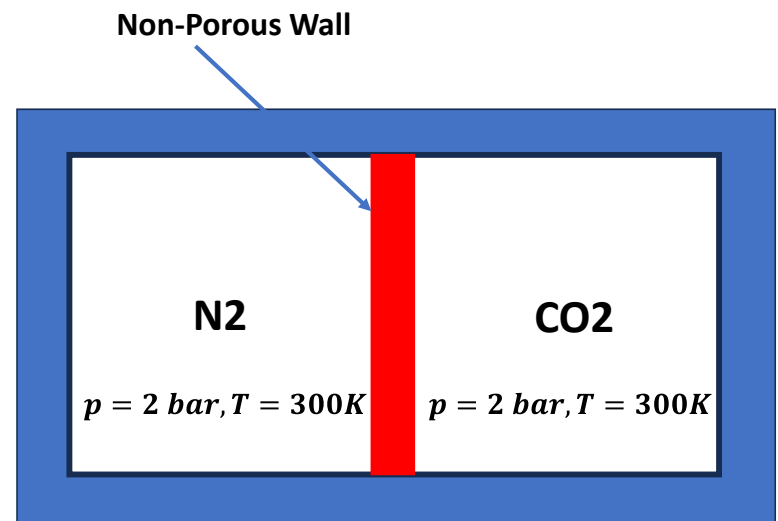
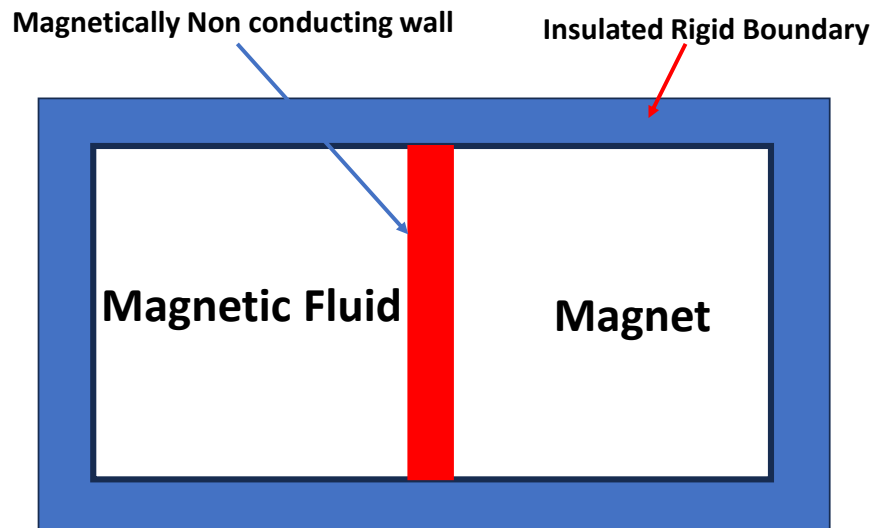
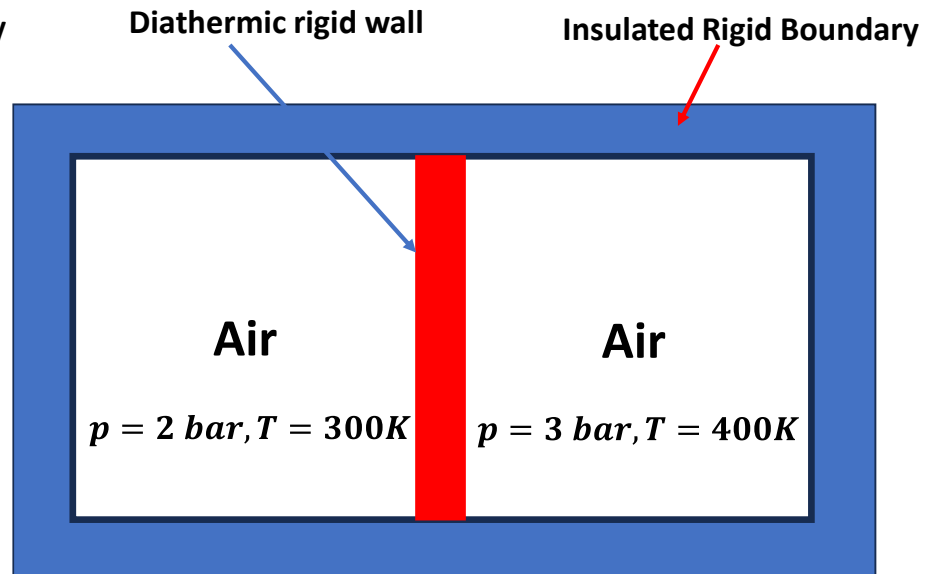
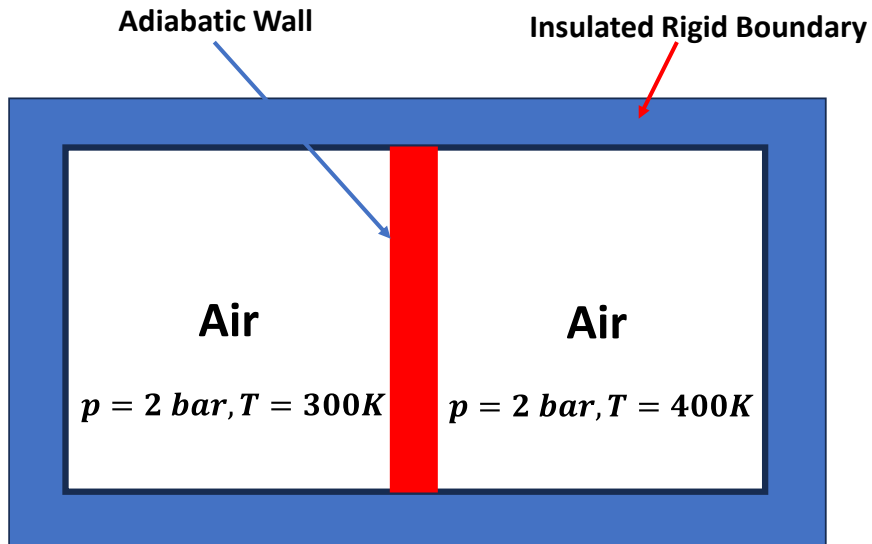
A ≡ ***U - TS*** → *Helmholtz Function*, ***a*** ≡ ***u - Ts*** → *Specific Helmholtz Function* = $\frac{A}{m}$

- ✓ The **thermodynamic equation of state** (EOS) is a formula relating the dependent & independent properties of the system. The ideal gas equation of state ***PV = mRT*** is one of them
- ✓ The thermodynamic **state** of system is the condition of the system described by its property. **When we are specifying the state the system must be in thermodynamic equilibrium.**

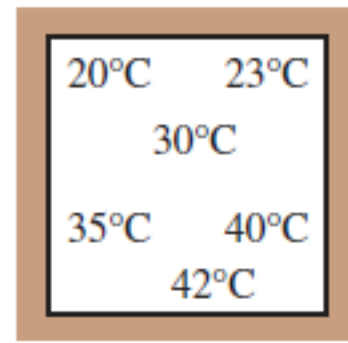
Thermodynamic Equilibrium

As already mentioned thermodynamics deals with equilibrium states and a system is said to be in thermodynamic equilibrium if it satisfies following equilibrium conditions. The state of thermodynamic equilibrium can be described in terms of macroscopic (thermodynamic coordinates) that do not involve the time. The condition that an isolated system attains when there are no internal constraints like adiabatic, rigid partitions, non-porous or electrically and magnetically non-conducting walls which prevents energy or mass exchange within the system is said to be one of the thermodynamic equilibrium. A system in such condition satisfies the following requirements

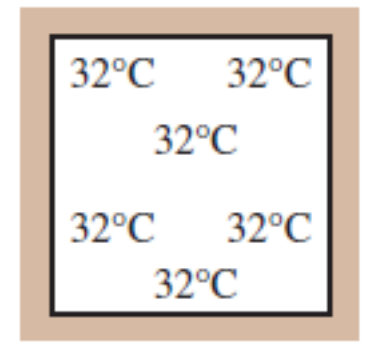
- Thermal equilibrium (related to property T)
- Mechanical equilibrium (related to force & torque)
- Phase and chemical equilibrium (related to property μ (chemical potential))
- Electrostatic equilibrium (Balance of charge throughout the system)



Thermal equilibrium (No energy transfer as heat occurs from point to point within the system. Temperature is uniform and invariable with time)

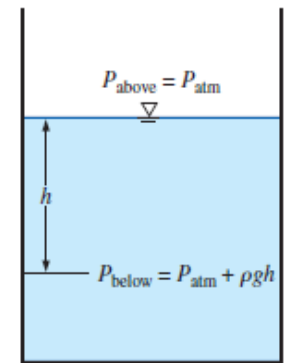


(a) Before



(b) After

Mechanical equilibrium (There should not be any unbalanced force or torque acting on any part of the system or the system as a whole caused by mechanical, electrical, magnetic and other interactions however, pressure may be different at different point) Imp. If the hydrostatic pressure variation is neglected then the pressure is uniform and invariable with time



Mechanical Equilibrium
Although $\Delta p \neq 0$

When there is an unbalanced torque and force in the interior of the system or between the system and surrounding the following phenomena takes place: turbulence, acceleration, eddies and waves.

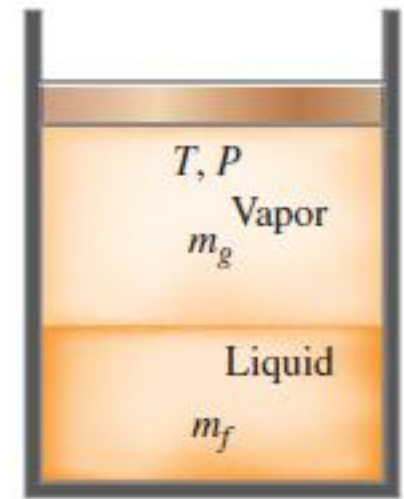
Phase: Chemically homogeneous and physically distinct part of matter. Any homogeneous part of the universe is said to be a phase

Phase Equilibrium: (The mass of individual phase remain unchanged) The chemical potential of both the phases must be same. (i.e. All the phases have same values of the specific Gibbs function)

Mathematically:

$$g_{Ph-1} = g_{Ph-2}$$

e.g. Suppose in an isolated chamber saturated liquid water and saturated vapour are in phase equilibrium, then the rate of condensation of steam must be equal to rate of vaporization of saturated liquid water, hence the phase equilibrium is some kind of dynamic equilibrium



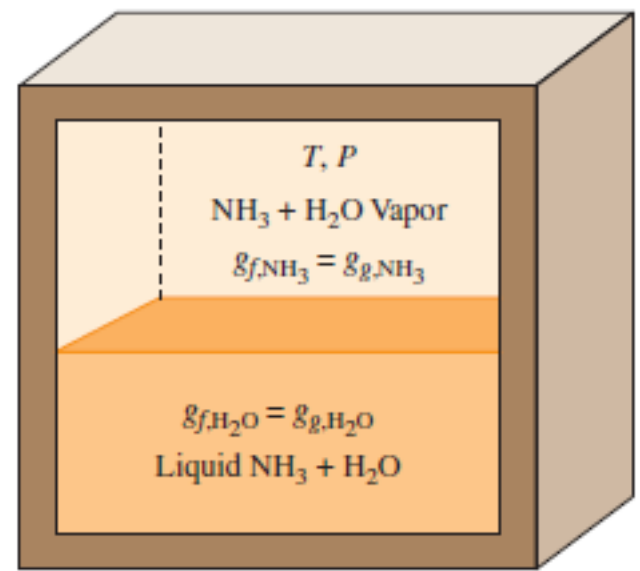
A liquid–vapor mixture in equilibrium at a constant temperature and pressure.

Many **multicomponent** systems encountered in practice involve two or more components . A **multicomponent multiphase system** at a **specified temperature and pressure** is in **phase equilibrium** when there is no driving force between the different phases of each component. Thus for phase equilibrium the **specific Gibbs function** ($g = h - Ts$) of each **component must be the same in each phase** i.e.

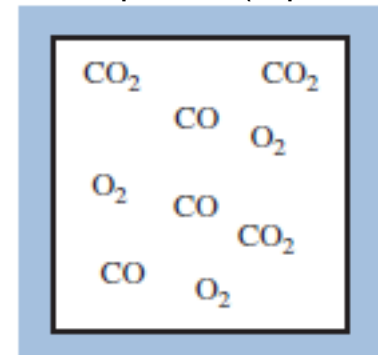
$$g_{f,1} = g_{g,1} = g_{s,1} \text{ for component 1}$$

$$g_{f,2} = g_{g,2} = g_{s,2} \text{ for component 2}$$

$$g_{f,N} = g_{g,N} = g_{s,N} \text{ for component } N$$



The figure shows a two component (NH_3 and H_2O) and two phase (liquid & vapour)

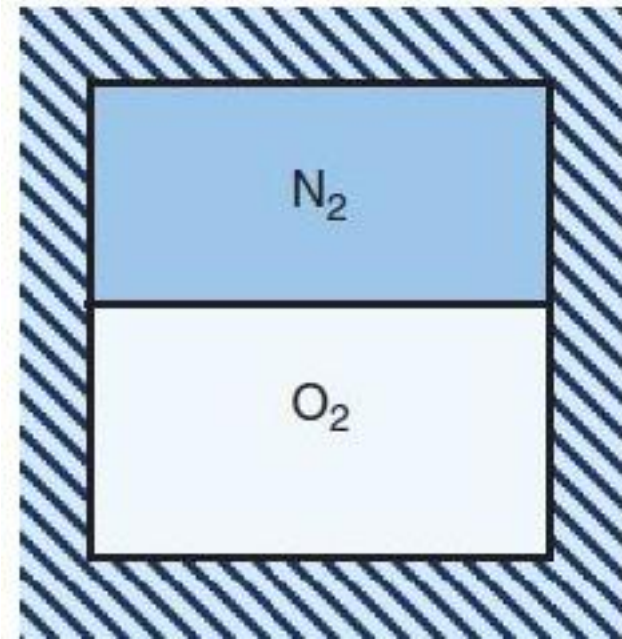


A reaction chamber that contains a mixture of CO_2 , CO , and O_2 at a specified temperature and pressure.

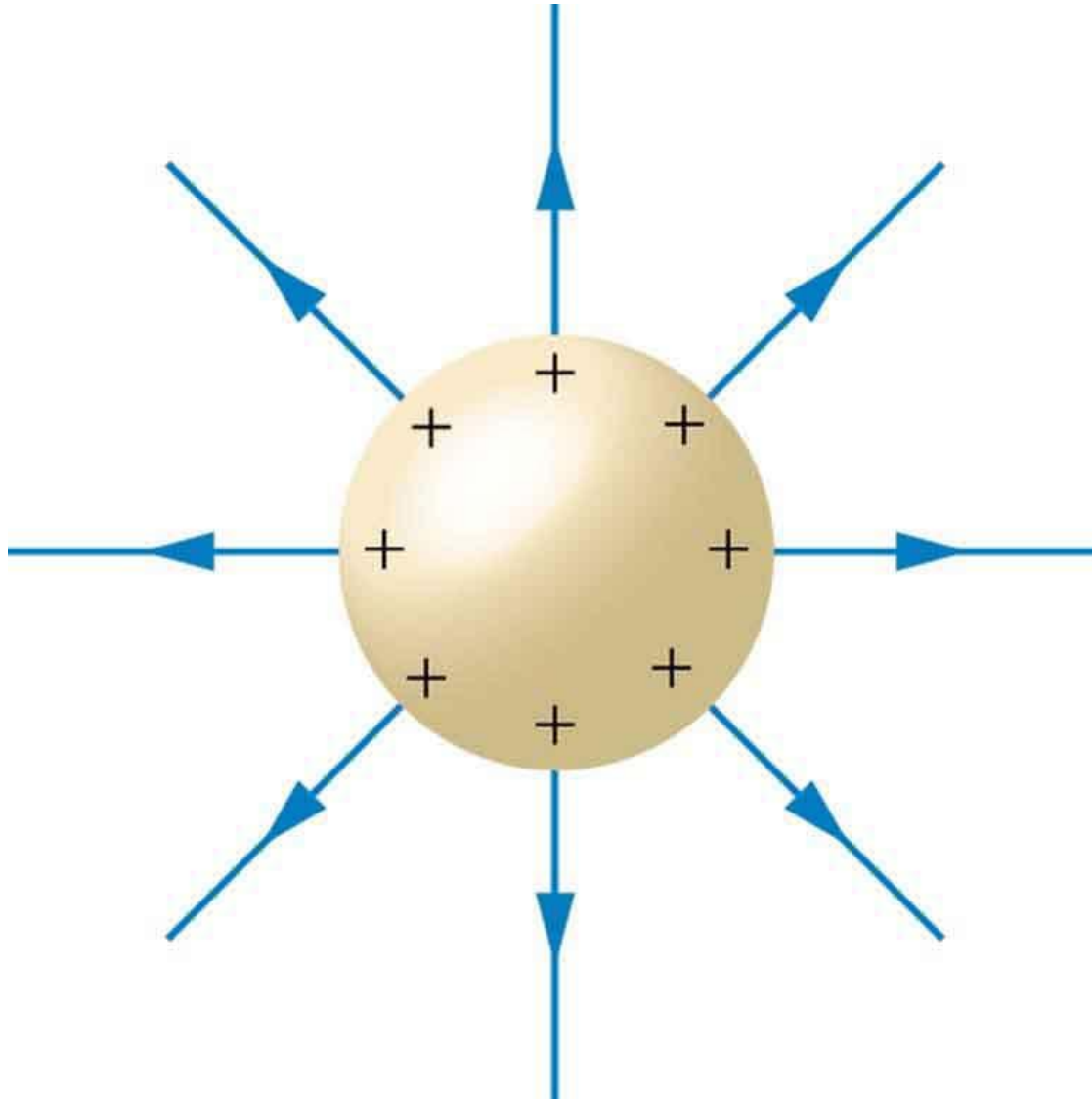
Chemical Equilibrium: (There should not be any potential for chemical reaction to occur or for the motion of any chemical constituent from one part of the system to the other part such as diffusion or solution)

Other aspect of Chemical Equilibrium

Consider the container at uniform temperature and pressure as shown in the figure, divided by a partition into two parts, one containing nitrogen and other oxygen. A small hole in the partition allows the flow of the gases from one part to other. Here the system is in mechanical and thermal equilibrium, but not in thermodynamic equilibrium due to lack of chemical equilibrium. μ_{N_2} and μ_{O_2} are not uniform throughout the system. Some changes within the system (Diffusion of O_2 and N_2 from higher concentration to lower concentration) will take place without any interaction with the environment and will continue until a uniform composition in the whole system is reached.



Electrostatic equilibrium: Is a state where all free electrical charges have stopped moving, and the charge distribution is fixed.



Temperature, Thermal Equilibrium, Zeroth Law

Though everybody is familiar with the property **Temperature** , defining it in words is extremely difficult

Though temperature is related to **the sense of hotness or coldness**, relying purely on **sensory perception could be misleading** as different objects (e.g. metal and wood) both at same temperature gives different sensation of coldness/hotness

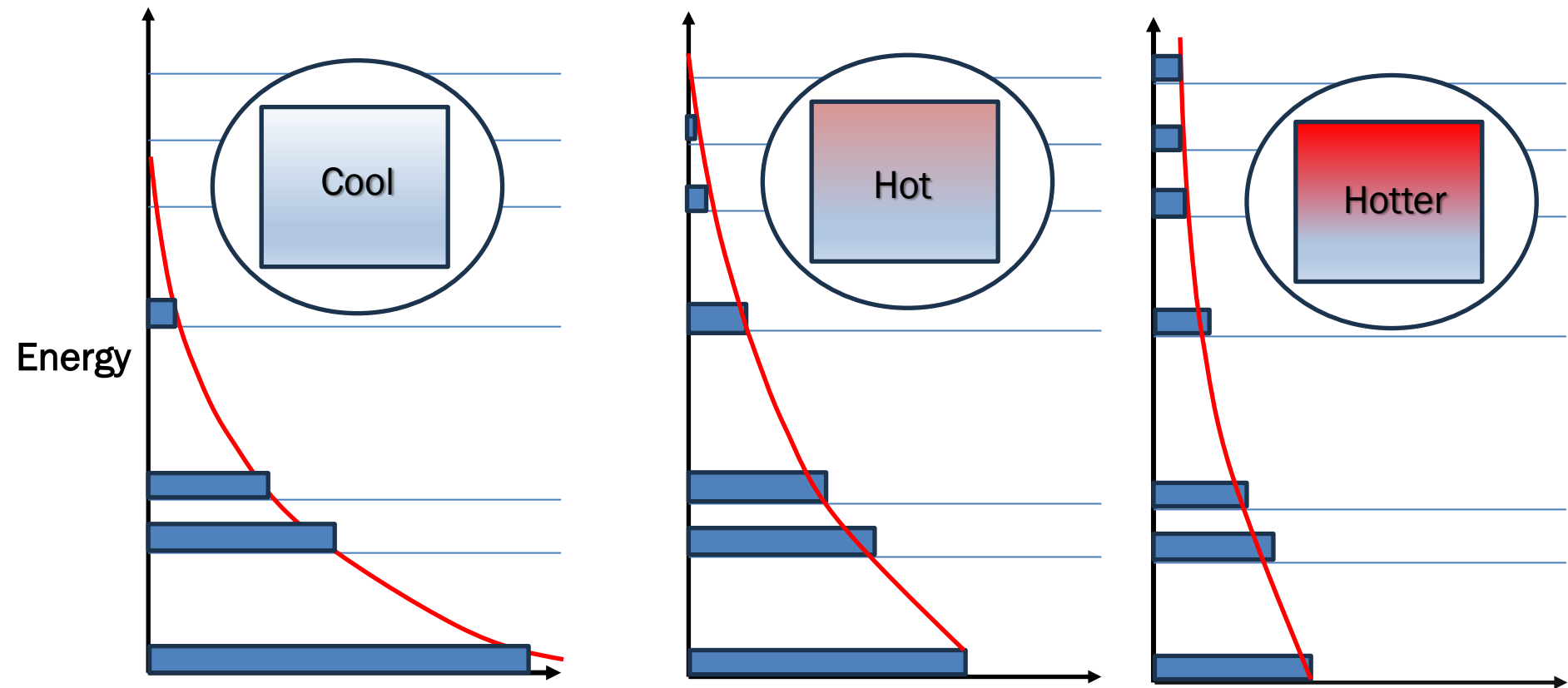
Due to difficulty in defining temperature in absolute terms an **equality of temperature** is defined

Two systems are said to have same temperature when brought in contact through a **diathermic (from Greek words for “through” and “warm”) wall** no change in any observable property occur when equilibrium condition.

Temperature is the parameter that **tells us the most probable distribution of the populations of molecules over available states of a system at equilibrium.**

$$\frac{\text{Population of state of energy } E}{\text{Population of state of energy } 0} = e^{-\beta E} \quad \beta = \frac{1}{kT}$$

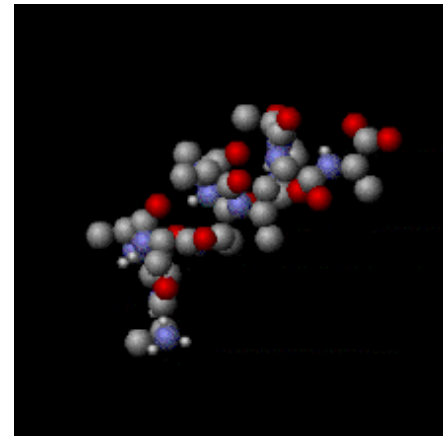
When temperature is **high many states have significant population**; when temperature is low only the states closed to the lowest state have significant population



$$\frac{\text{Population of state of energy } E}{\text{Population of state of energy } 0} = e^{-\beta E}$$

We can think of temperature as an indication of the average speed of molecules in a gas

$$\bar{V} \propto \sqrt{T}$$

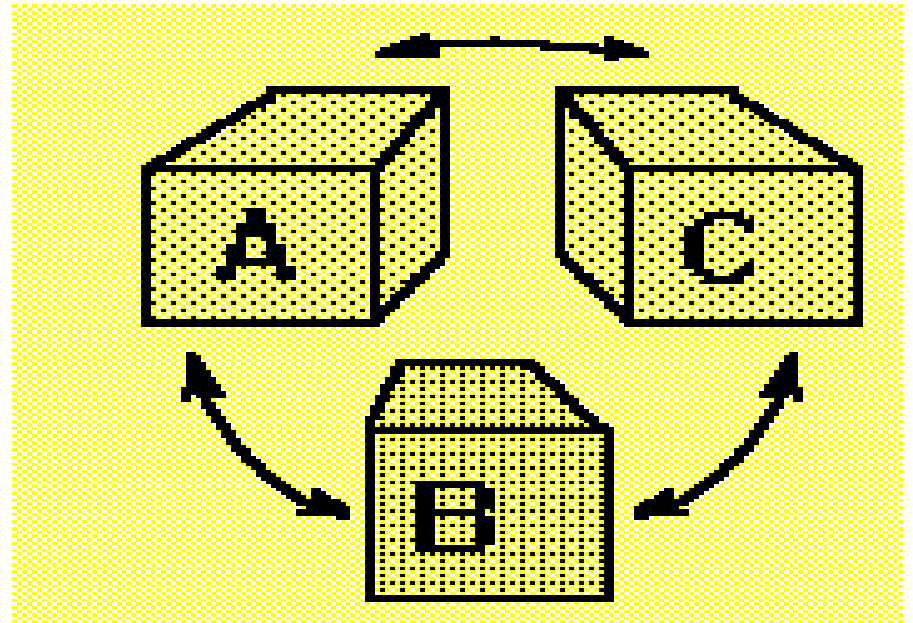
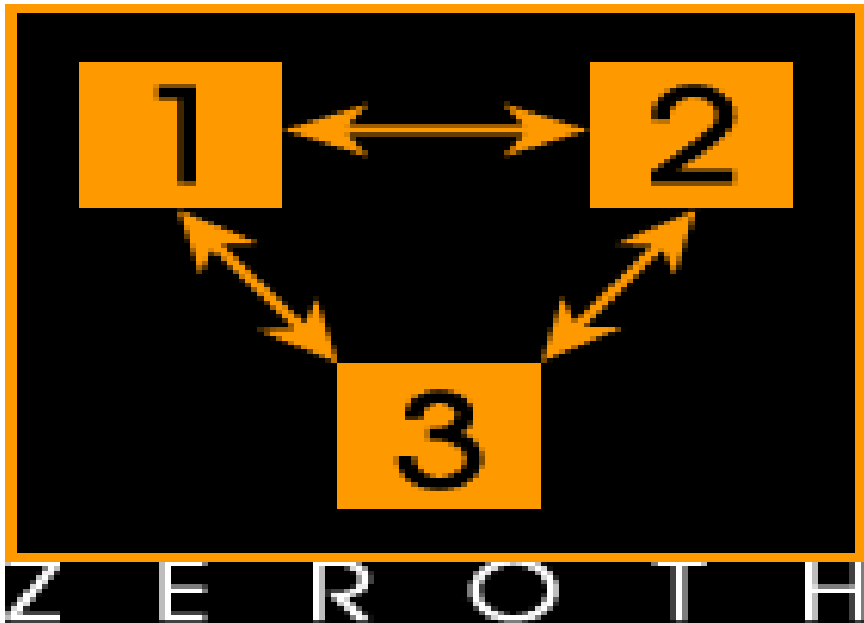


Mutual Equilibrium

- ❑ Consider **two systems**, each in equilibrium. When brought into contact, the systems may or may not interact depending on their states. If no interaction takes place, the systems are said to be in mutual equilibrium.
- ❑ Another way of looking at it is that mutual equilibrium exists if **a combined system composed of these two systems is in a state of equilibrium**.
- ❑ It should be noted that each of the two systems may be in equilibrium, yet not in mutual equilibrium. In that case, when the systems are brought into contact, an interaction will take place. The interaction will continue until mutual equilibrium is reached.

Zeroth law of thermodynamics: The concept of equilibrium may now be used to formulate the zeroth law of thermodynamics. *Two systems, each in mutual equilibrium with a third system, are also in mutual equilibrium with each other.*

Zeroth law of thermodynamics: Talks about **thermal equilibrium** and **temperature**. The property temperature, **springs from the zeroth Law**. If **two** bodies A and B are in thermal equilibrium with body C, then A and B must be in thermal equilibrium with each other irrespective of whether they are in direct contact with each other or not



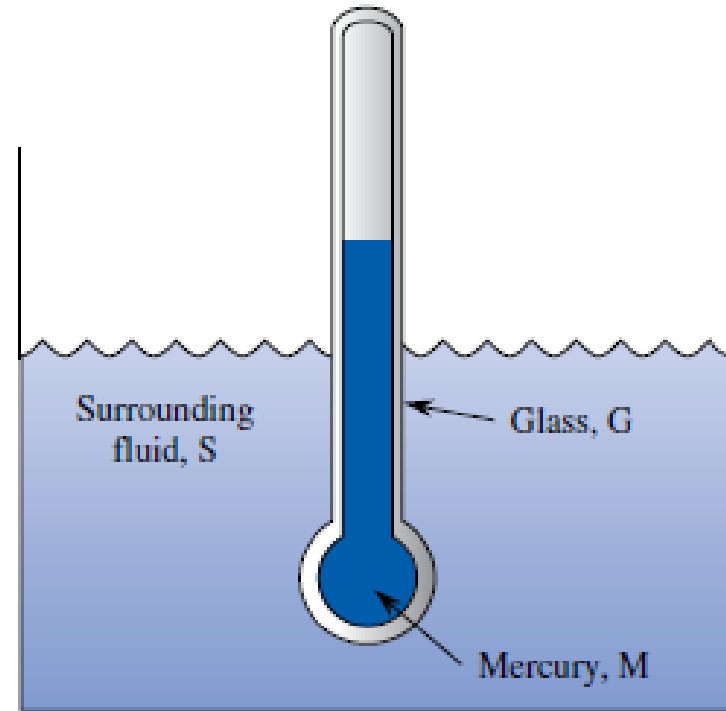
❖ This may seem trivial at first reading, but consider this: if a man A loves a woman C and man B loves woman C does it follow that man A loves man B?

❖ For example, two substances (such as hydrogen and oxygen) which are inert with respect to a third (for example argon) need not be chemically inert with respect to each other. A mixture of H_2 and O_2 explode violently when ignited, but a mixture of either of them with argon will not.

❖ Zeroth law forms the theoretical basis for temperature measurement

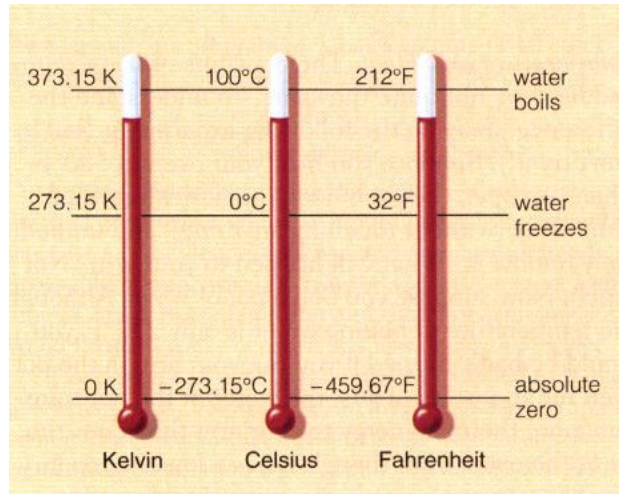
All systems in thermal equilibrium with each other have a common property

❖ The Zeroth law tells us that in case of mercury in glass thermometer if the glass is at the same temperature as (i.e., is in thermal equilibrium with) that of the surrounding fluid and if the mercury is at the same temperature as the glass, then mercury is at the same temperature as the surrounding fluid. Thus the thermometer can be graduated to show the mercury temperature and this temperature will automatically (via the zeroth law) be equal to the temperature of the surroundings.

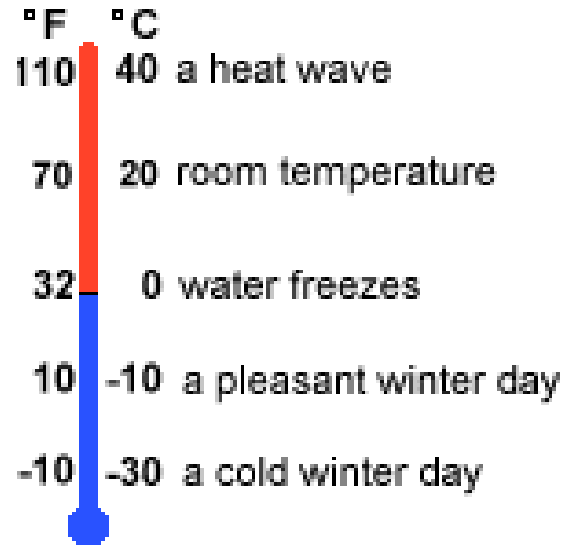


Zeroth law;
 $T_G = T_S$
and
 $T_M = T_G$
therefore,
 $T_M = T_S$

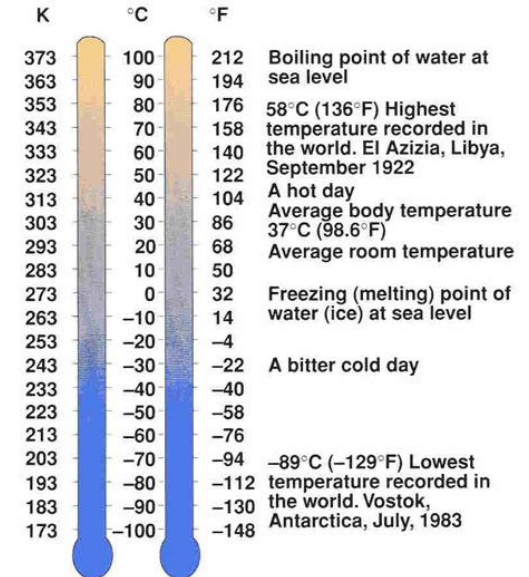
Temperature Scales



Temperature Ranges



Temperature Ranges



There are two absolute scale related as follows

- Rankine Scale
- Kelvin Scale

$$T(R) = \frac{9}{5} T(K)$$

There are two relative temperature scales associated with the above absolute scales

$$T(F) = \frac{9}{5} T(^{\circ}C) + 32$$

$$T(R) = T(^{\circ}F) + 459.67$$

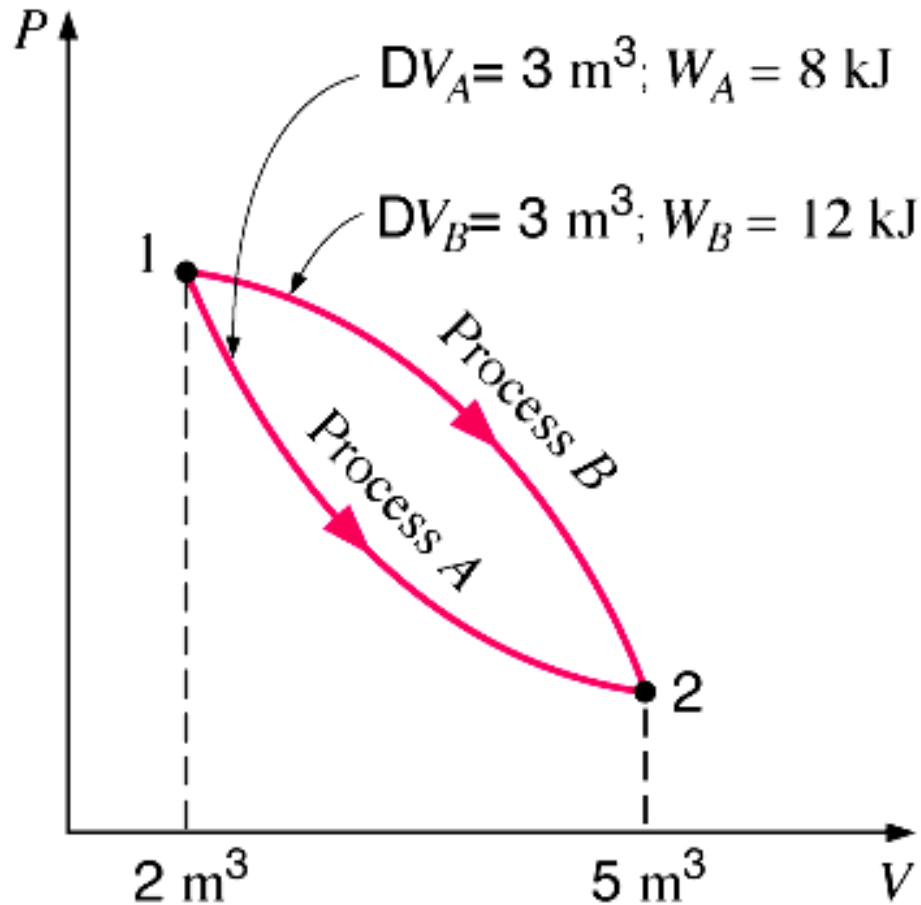
$$T(K) = T(^{\circ}C) + 273.15$$

$$\frac{C - 0}{100 - 0} = \frac{F - 32}{212 - 32} = \frac{K - 273}{373 - 273} = \frac{X - LFP}{UFP - LFP}$$

We often encounter situations in which the system changes continuously but still goes through states of equilibrium at any instant of time. That, of course, is not contradictory to the concept of equilibrium as those changes in the system take place while corresponding changes take place in the environment. The following example may illustrate this point. Consider a gas enclosed in a cylinder–piston assembly for which all the intensive properties, such as pressure, temperature, density, and so on, are uniform. This system is in a state of equilibrium regardless of whether the pressure in the environment is higher, lower, or equal to that inside the cylinder. If the pressures outside and inside the cylinder are equal, change will take place neither in the system nor in the environment, and the situation is easy to associate with equilibrium. Even if the pressures are not equal and the piston moves, the gas may still go through equilibrium states as long as the intensive properties of the gas in the cylinder remain uniform at any instant. All that is needed to ascertain that the system is in a state of equilibrium is to check whether changes would continue once we stop the interaction with the environment, by stopping the piston. In that case if the state of the gas could not be changed, then it is in equilibrium.

Thermodynamic Process & Cycle

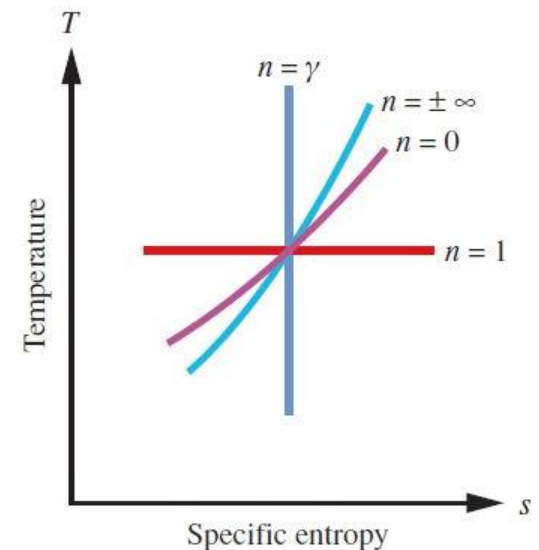
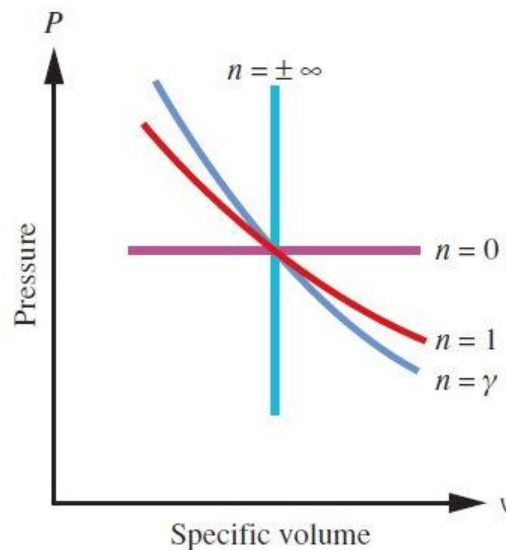
A **process** is defined as the path of thermodynamic states which the system passes through as it goes from an initial state to a final state. Understanding the nature of the process path is very important as heat and work depend on the path



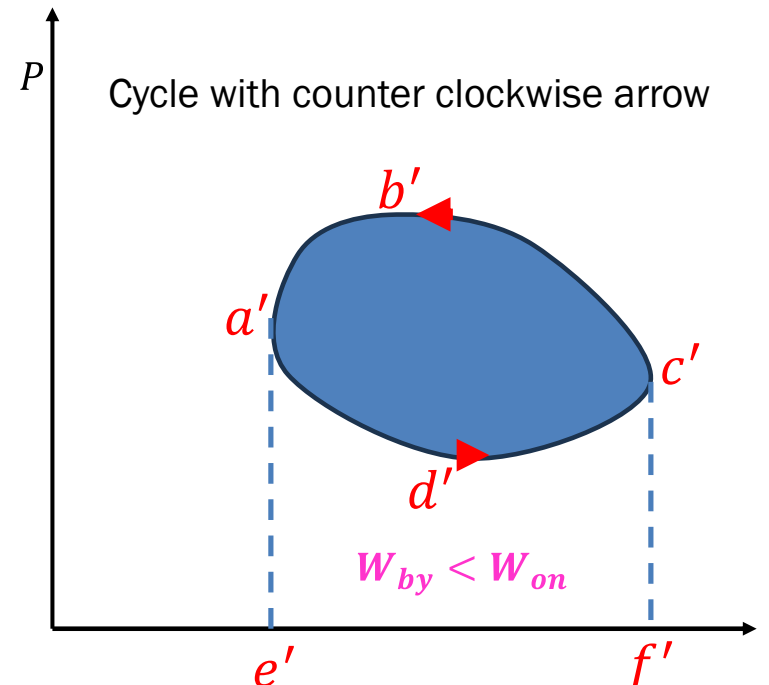
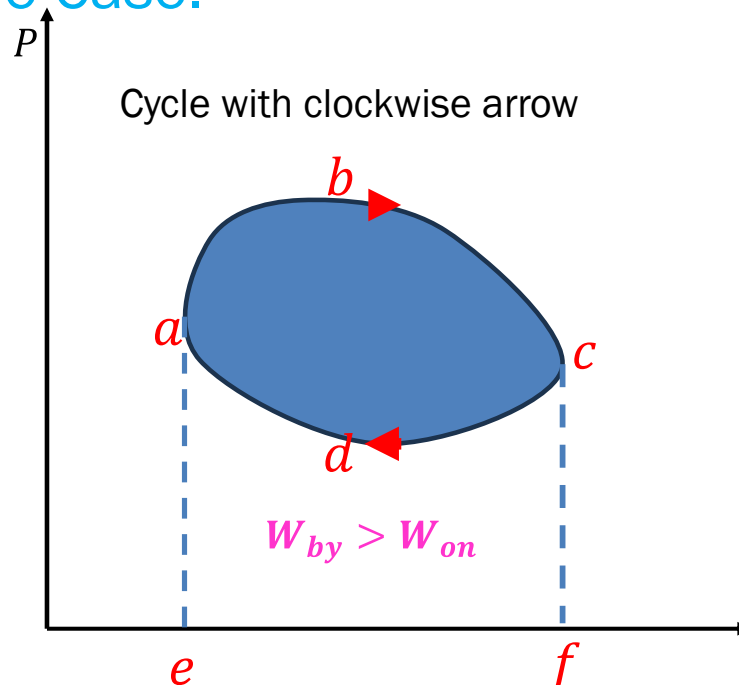
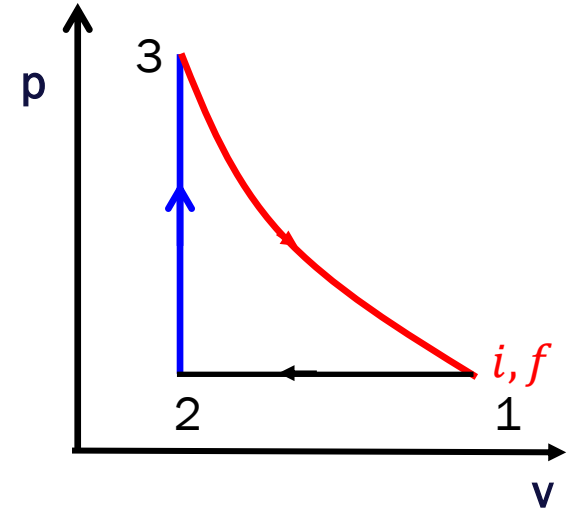
Areas under the curve A and B above the V axis are different (The path functions are somehow related to this area)

Common Thermodynamic Process

1. Constant Volume Process → *Isochoric*
2. Constant Pressure Process → *Isobaric*
3. Constant Temperature Process → *Isothermal*
4. Constant Enthalpy Process → *Isenthalpic*
5. Constant Entropy Process → *Isentropic*
6. Polytropic Process → $PV^n = \text{const}$; Hyperbolic Process → $PV = \text{constant}$
7. Adiabatic Process → $\delta Q = 0$
8. Aergonic Process → $\delta W = 0$



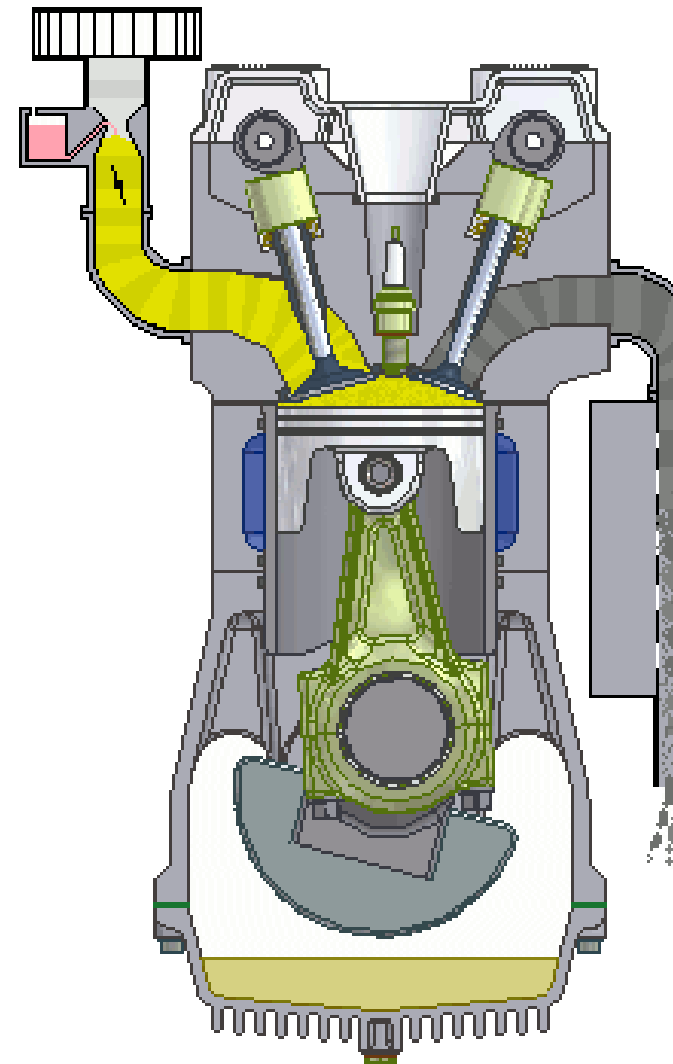
A system is said to have **undergone a cycle** if beginning with an initial state it goes through sequences of different processes and finally arrives at the initial state. To get **continuous effect** from any system, it has to operate on a **thermodynamic cycle**. The area bounded by the closed curve gives the net work or net heat interactions and the arrow mark will decide the direction of net work and net heat for reversible Case.



A **mechanical cycle** (all mechanical components begin and end with same configuration) is **different from thermodynamic cycle** (working fluid begin and end with the same condition)

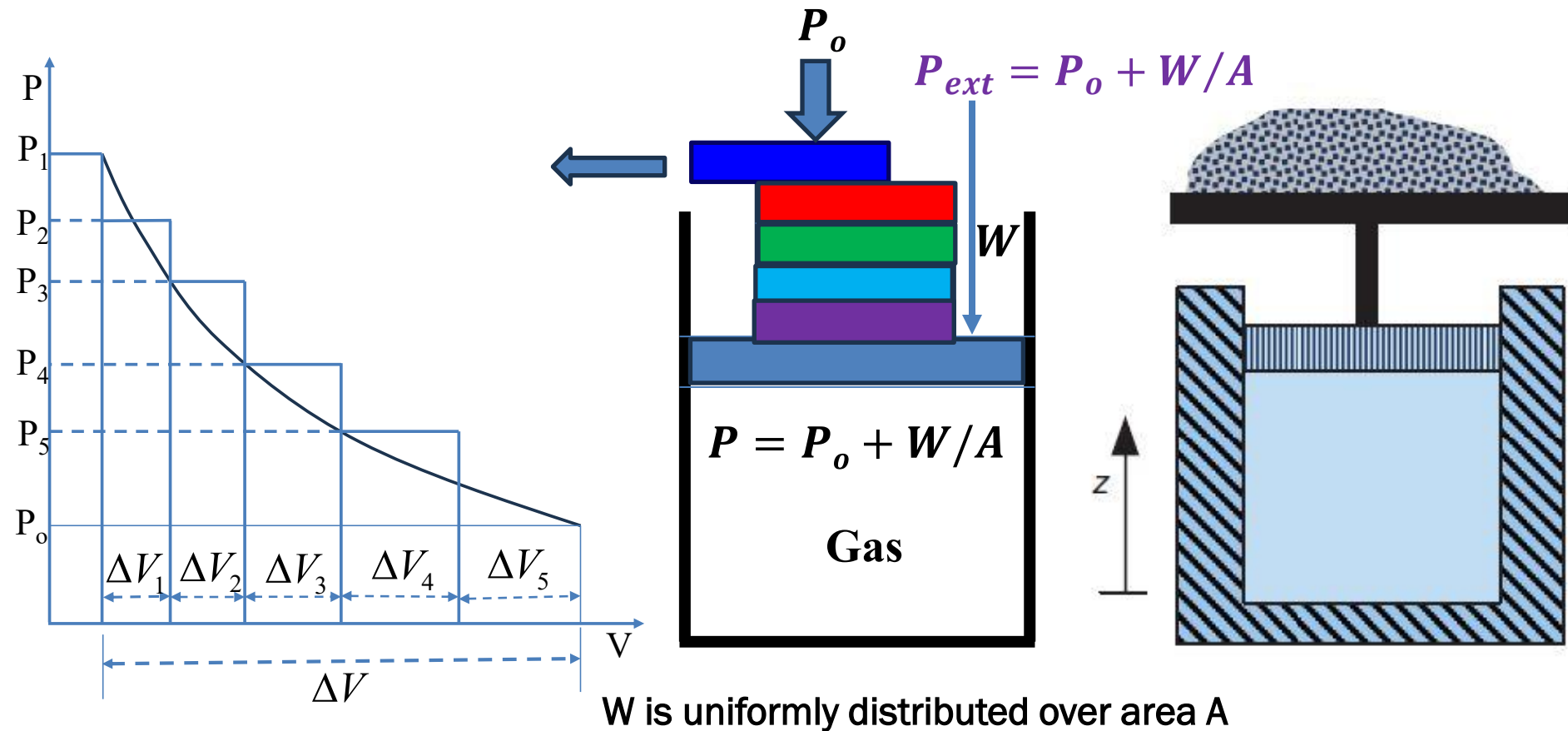
Example: A 4-stroke petrol engine – two **mechanical cycle** for 2 crank shaft rotation.

But the working fluid changes from **fuel + air** to **combustion product**. For this system to under go a thermodynamic cycle the **exhaust gas** have to be converted back into fuel and air

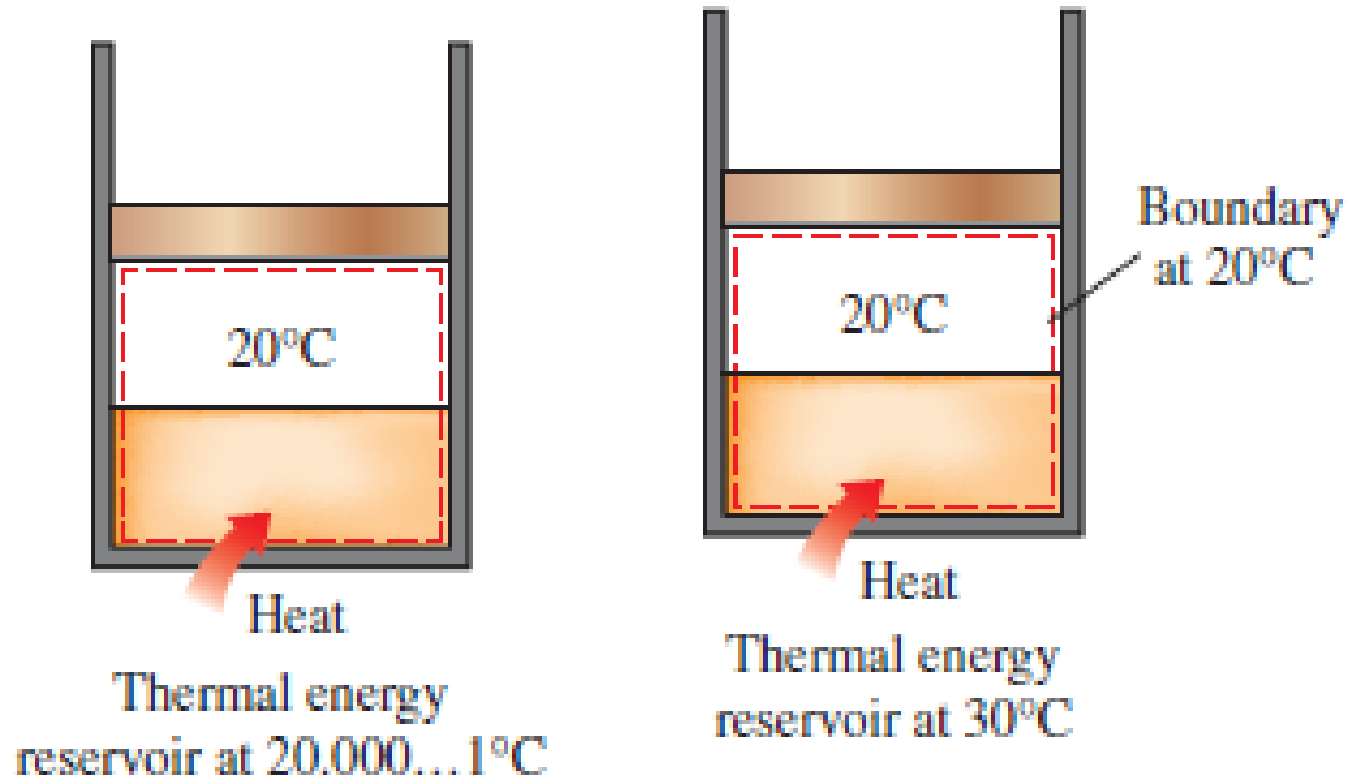


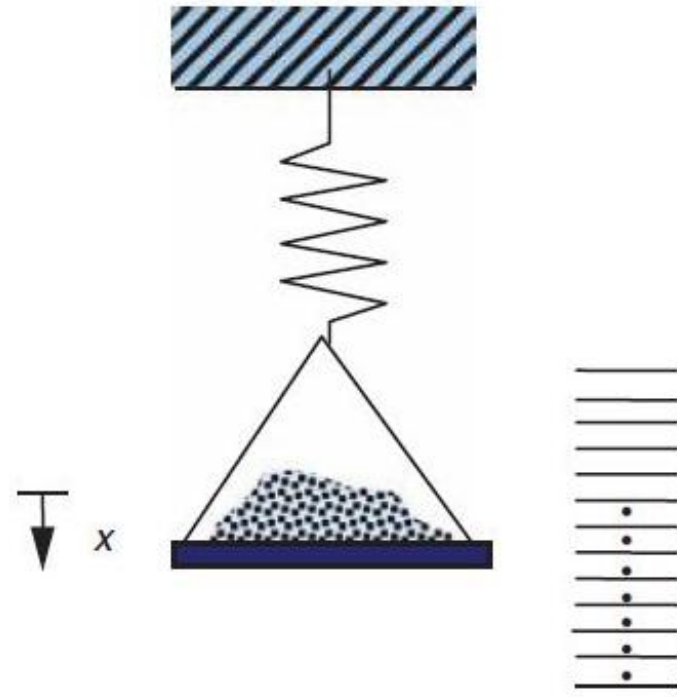
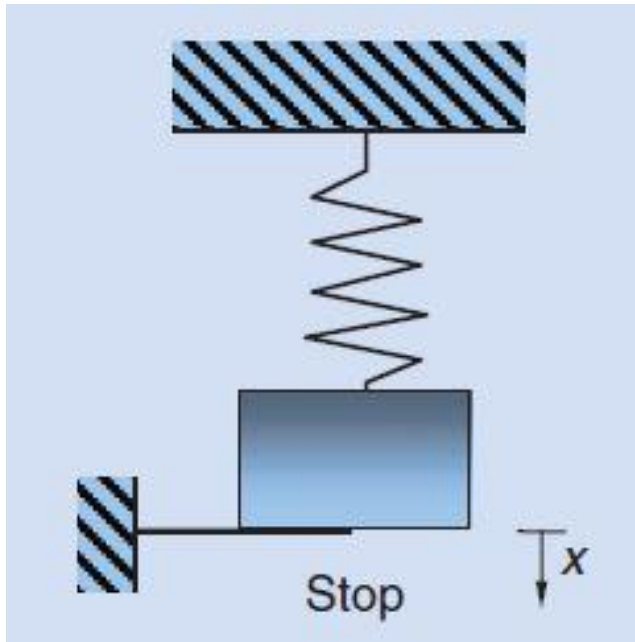
Quasi-Static Process

- ❖ During a quasi-static (*very slow*) process, the system is at all time *infinitesimally near a state of thermodynamic equilibrium*, and all the state through which the system passes can be described by means of thermodynamic coordinates referring to the system as a whole.



- ❖ An equation of state is valid for all these state. The variables in an equation of state are essentially the same in both the system and surroundings.
- ❖ During quasi-static process with proper boundary. (the temperature is the same on both sides of a diathermic wall, but not for an adiabatic wall)
- ❖ A quasi-static process is an idealization applicable to all thermodynamic systems, including electric and magnetic systems.





- ❖ At this point , *a quasi-static process is synonymous with reversible process but a quasi-static process can be irreversible when friction is taken into account.*

Fixing the State

It is known that the state can be uniquely determined by giving the values of independent properties. All other properties are determined from this independent subset. A general **state principle** has been developed as a guide in **determining the number of independent properties** required to fix the state of a system. **The state principle says about the intensive state of a system**

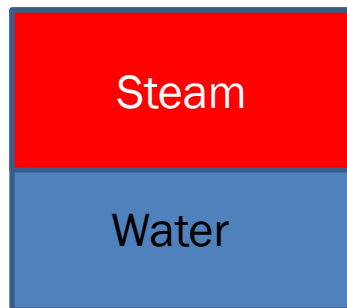
- For a **pure, simple compressible system** the state postulate indicates that **two independent intensive properties** are required to fix the intensive state of the system. In other words from these two properties other intensive properties can be calculated but **total values can only be calculated if mass is known.**

Properties Of Pure Substance

Any system uses a working substance (may be pure substance /multi component mixture etc.) for energy transfer or transformation.

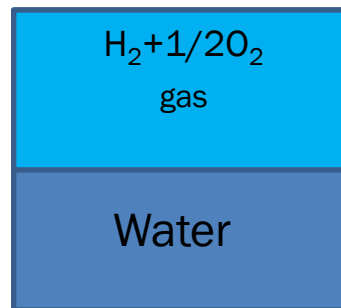
Pure substance:

1. Homogeneous in chemical composition
2. Invariable in chemical composition
3. Homogeneous in chemical aggregation
4. Invariable in chemical aggregation
5. It may have more than one phase



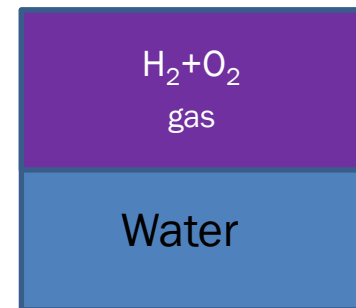
1&2

System-1



1

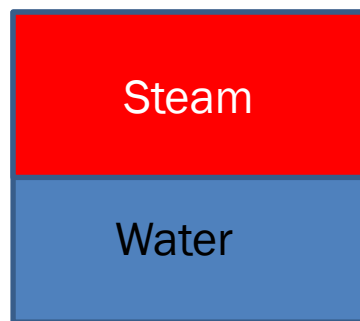
System-2



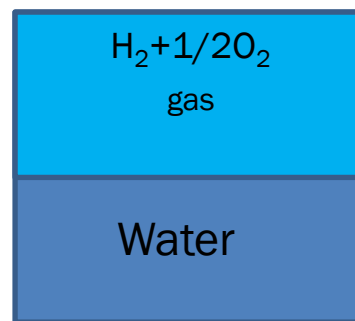
None

System-3

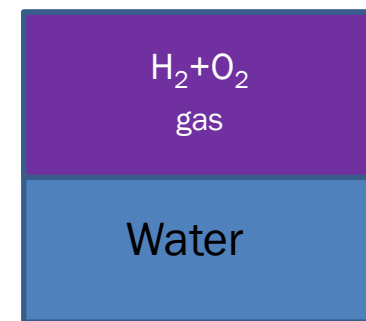
“Homogeneous in composition” means that the composition of each part of the system is the same as the composition of every other part. “Composition” means the relative proportions of the chemical elements into which the sample can be analyzed. It does not matter how these elements are combined. As shown in system 1 both steam and water have the same composition ($\text{H}:\text{O}=2:1$), similarly in system-2 both the gas mixture and water have the same composition ($\text{H}:\text{O}=2:1$), however in system-3 the composition is inhomogeneous gas mixture ($\text{H}:\text{O}=1:1$), Water ($\text{H}:\text{O}=2:1$),



System-1



System-2



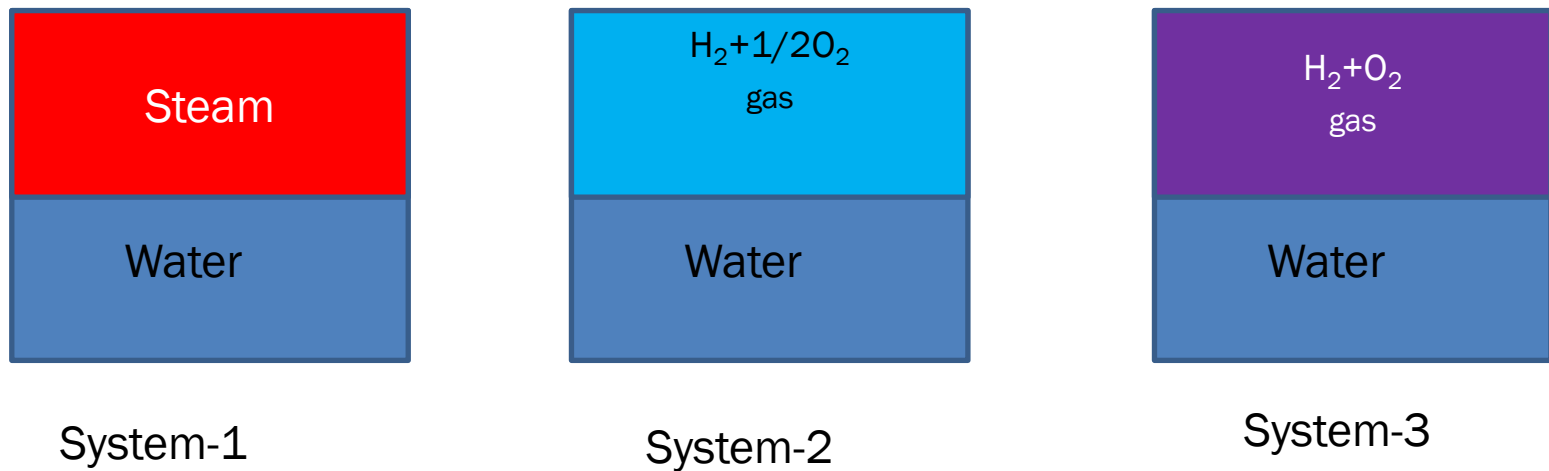
System-3

Both the system-1 & 2 satisfy the condition 1&2 but System-3 does not

The system-1 satisfies condition 3&4 but system2 &3 do not, hence System-1 is pure substance

“Homogeneous in chemical aggregation” means that the chemical elements must be combined chemically in the same way in all parts of the system

As shown in system-1 both steam and water have the same Chemical aggregation i.e. Hydrogen and Oxygen are bonded similarly both in steam and water, However in system-2 and 3 the bonding between Hydrogen and Oxygen in gas mixture and water are different

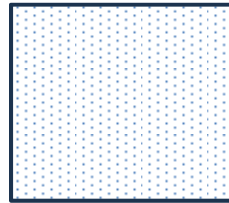


“In Variable in chemical aggregation ” means the state of the chemical combination of the system does not change with time

Homogeneous (Single Phase)

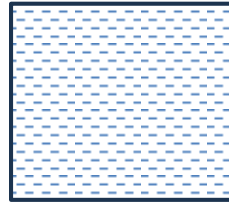
Heterogeneous (multiphase)

Singlecomponent



Pure substance

Steam (H_2O)

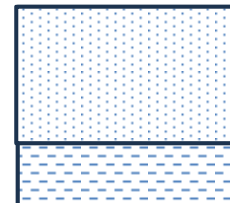


Water (H_2O)

Simple Substance

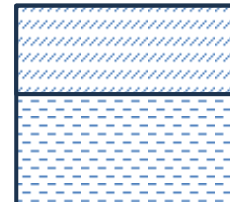


Ice (H_2O)

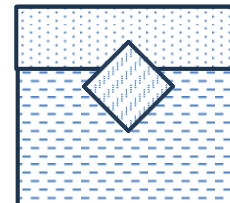


Pure substance

Steam & water

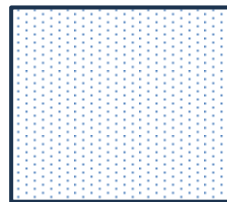


Ice & water



Steam, Water and Ice

Multicomponent

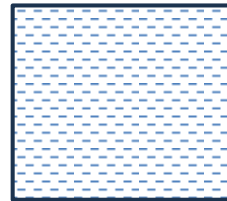


Pure substance

Air (Gaseous mixture)
(N_2 , O_2 , CO_2 etc.)

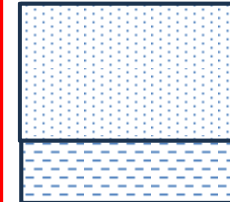


Product of Combustion
gaseous mixture: CO_2 ,
 H_2O , N_2 , O_2)



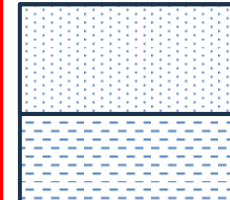
Sea water (Liquid
mixture: H_2O , $NaCl$,
etc.)

Simple Substance

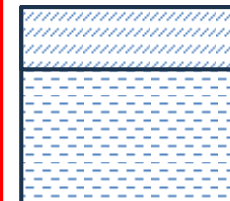


Not a Pure substance

Gaseous air & Liquid Air



Gaseous mixture
(fuel vapor, air)
and liquid fuel



Sea ice and
sea water

1. Is moist air a pure substance ?

It is well known that moist air is a homogeneous mixture of different **permanent gases** (O_2 , N_2 , CO_2 , Ar, etc.) and **water vapour**

Dry air :

Constituent	Molecular weight	Mol fraction
Oxygen	32.000	0.2095
Nitrogen	28.016	0.7809
Argon	39.944	0.0093
Carbon dioxide	44.010	0.0003

The **dry air** & **water vapour** both are **pure substances**, however the water vapour part of moist air may change in any air conditioning process, hence **moist air is not a pure substance**, rather a **binary mixture** of dry air and water vapour.



- **Moist air is less dense compared to the dry air.** $M_{H_2O} = 18 \text{ kg/kmol}$

2. Is mixture of liquid air a gaseous air pure substance ?

- At standard atmospheric pressure and temperature, the dry air is a mixture of approximately 78.09% N_2 , 20.95% O_2 , 0.93% Ar, 0.04% CO_2 , and other gases in meagre amounts.
- The **gaseous** as well as **liquid air** are pure substances, however, the **mixture of gaseous and liquid air is not a pure substance**. The concentration of N_2 in liquid air is higher than that in the gaseous air, then the chemical composition is no more homogeneous.



3. Is mixture of ice, water and steam is a pure substance ?

As the chemical composition as well as aggregation in all these three phases of water is same, the mixture can be treated as pure substance

4. Is combustion products is a substance ?

In a combustion process the reactants (air and fuel vapor mixture and liquid fuel) is not a pure substance but the combustion product (homogeneous mixture of CO_2 NO_x CH_4 etc.) can be treated as a pure substance

Important characteristics of pure substance:

State Postulate: To specify the state of a simple compressible pure substance two independent intensive properties are required. Simple system means absence of the effect of electric, magnetic, anisotropic stress field (*pressure is isotropic in nature*), surface tension, gravity etc. There is only one mode of work interaction. If it is a compressible system that one mode is the moving boundary work. The number 2 came from the fact that 1 related to heat interaction and 1 is related to the single mode of work interaction.

One of the important property of a pure substance is that during the phase change process the pressure and temperature are dependent properties. So if you fix the pressure the temperature is automatically fixed during phase change (water boils at 100°C when the pressure is maintained at 1 bar). The azeotropic mixtures also behave like pure substance.

Pure Substance						Pure substance
	1	2	3	4	5	
Steam	✓	✓	✓	✓	1	✓
Steam & water	✓	✓	✓	✓	2	✓
Air (O2 +N2 gas)	✓	✓	✓	✓	1	✓
Air + liquid air	✗	✗	✓	✓	2	✗
Air + fuel vapor no reaction	✓	✓	✓	✓	1	✓
Air +fuel vapor & liquid fuel	✗	✗	✓	✓	2	✗
Air + fuel vapor & combustion product	✓	✓	✗	✗	2	✗
Gaseous Combustion Products	✓	✓	✓	✓	1	✓

- A. **Simple Compressible System:** The term simple system is applied when there is **only one way the system energy can be significantly altered by work interaction** as the system undergoes **quasiequilibrium** processes. The most important of these is the simple compressible system .
- B. As suggested by the name, the **change in volume can have significant influence on the energy of the simple compressible systems**. The only mode of energy transfer by work that can occur as a simple compressible system undergoes quasiequilibrium processes is associated with **volume change $\int p dv$** .
- C. When the influence of the earth's gravitational field is negligible. Pressure is uniform through out the system. The other types of the simple system are **simple elastic** and **simple magnetic systems**.
- D. A simple system compressible system is free from **gravitational, electric, magnetic, stress** and **surface tension** effects.

Why 2 ?

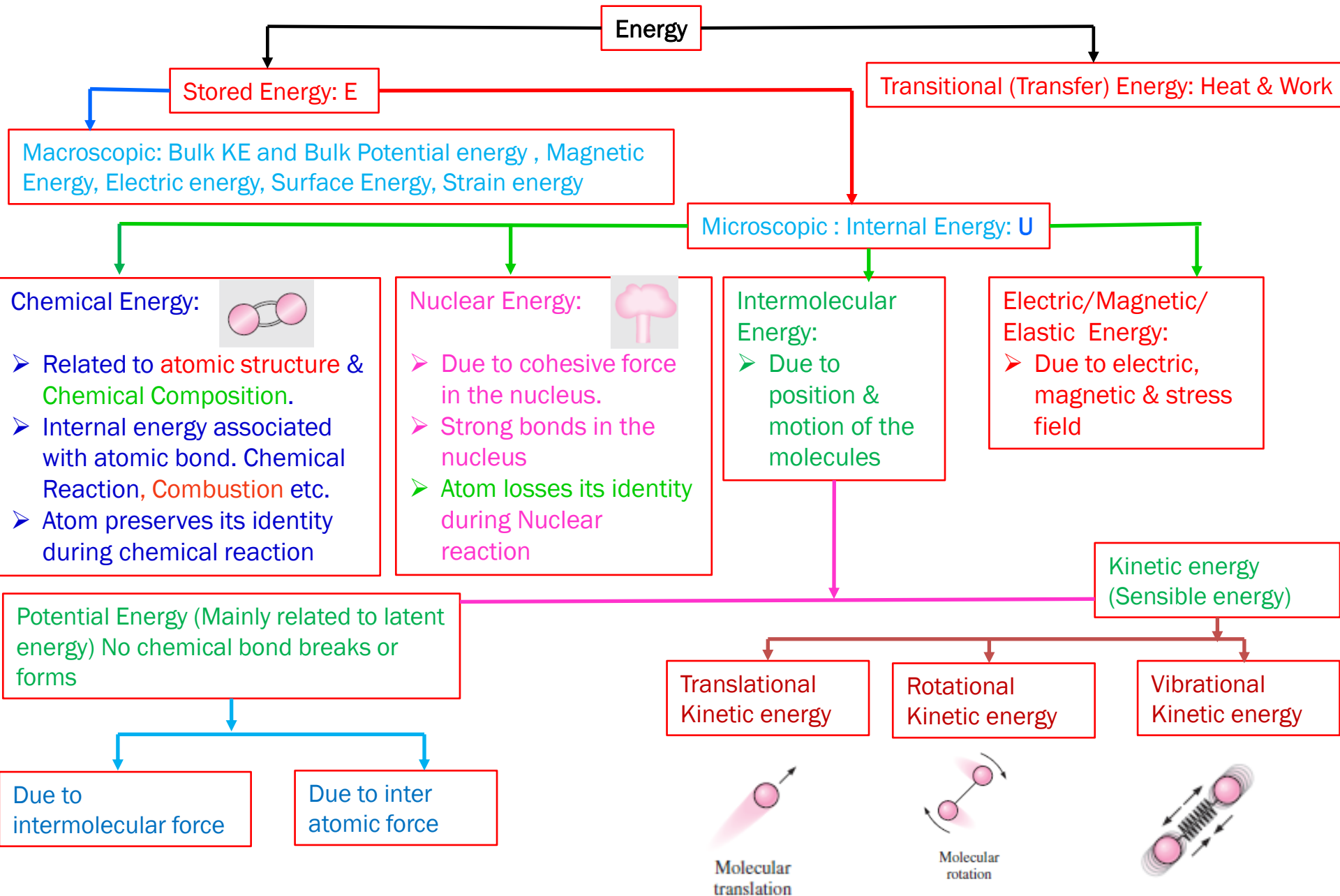
- **State principle:** Based on the empirical evidence, it has been concluded that there is **one independent property for each way a system's energy can be varied independently**.
- Energy of a **closed system** can vary by **heat and work**. Accordingly, one independent property can be associated with heat transfer as one way of varying the energy, and another independent property can be accounted for each relevant way the energy can be changed through work.
- On the basis of experimental evidence, therefore **the state principle asserts that the number of independent properties is one plus the number of relevant work interactions**. When counting the number of work, it suffices to consider only those that would be significant in “**Quasiequilibrium**” process of the system

Matter can be treated as having a property, called energy, that is an extensive, conserved scalar measuring its ability to cause change; work is a transfer of energy.

- extensive means that the energy of a system is the sum of the energies of its parts
- conserved means that the total amount of energy in an isolated system does not change;
- scalar means that energy is a quantity without directional (vector) character
- work is a transfer of energy; it provides the key to the quantitative evaluation of the energy contained by matter.

Energy: is the ability to produce an effect

Broad Classification of Energy



Bulk Kinetic Energy & **Bulk Potential energy** are the kinetic and potential energy of the **system/ control volume as a whole** measured **with respect to an external frame of reference**. (High grade energy)

$$\text{Stored Energy } (E) = U + \frac{1}{2}m\mathbf{V}^2 + mgz + ME + EE + SE + SDE$$

V is the **velocity of the center of mass** and **z** is the elevation of the center of mass of the system/control volume from the reference datum

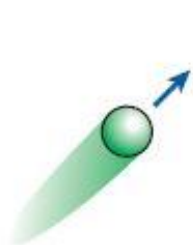
If a **compressible system** is affected by gravity its energy is a function of **elevation z** in addition to the two properties

$$E = f(\pi_1, \pi_2, z)$$

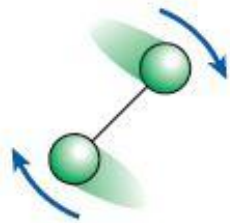
Molecular/atomic kinetic or potential energies are related to **motion and orientations** of the molecules and atoms with respect to an internal frame of reference. (**Low grade Energy**)

The **sensible** (**Molecular kinetic**) and **latent** (**Molecular Potential**) form of energy is generally known as Thermal Energy. (**Low grade Energy**)

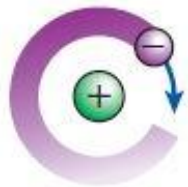
$$\textit{Stored Energy (E)} = U + \frac{1}{2}m\mathbf{V}^2 + mgz + ME + EE + SE + SDE$$



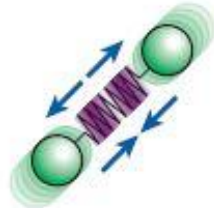
Molecular translation



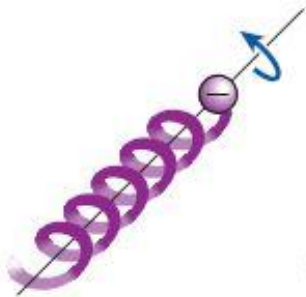
Molecular rotation



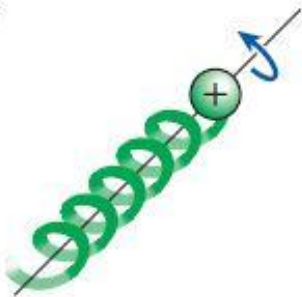
Electron translation



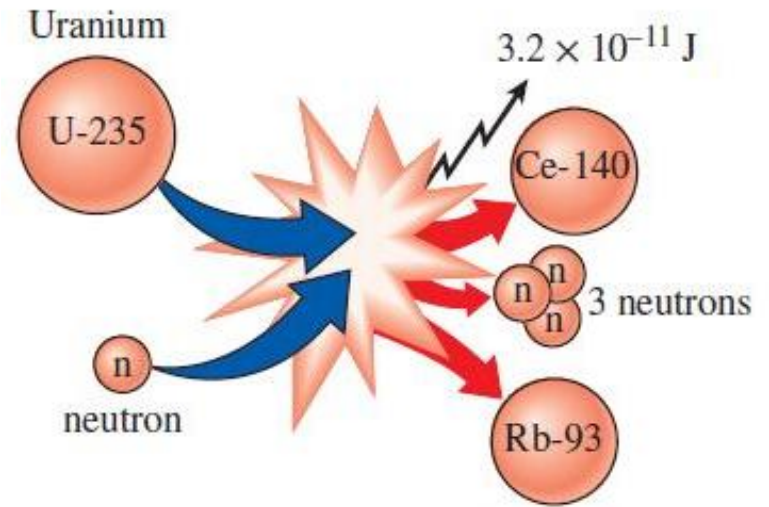
Molecular vibration



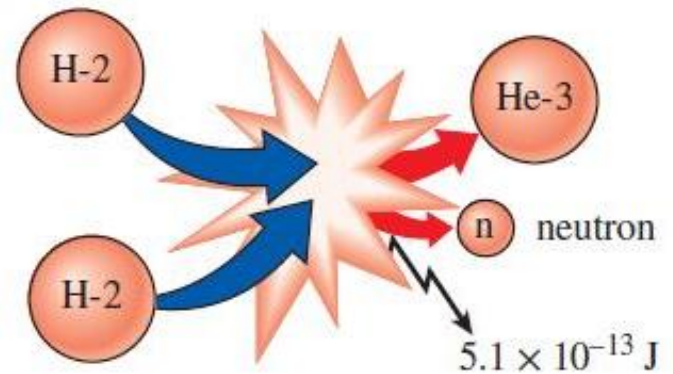
Electron spin



Nuclear spin

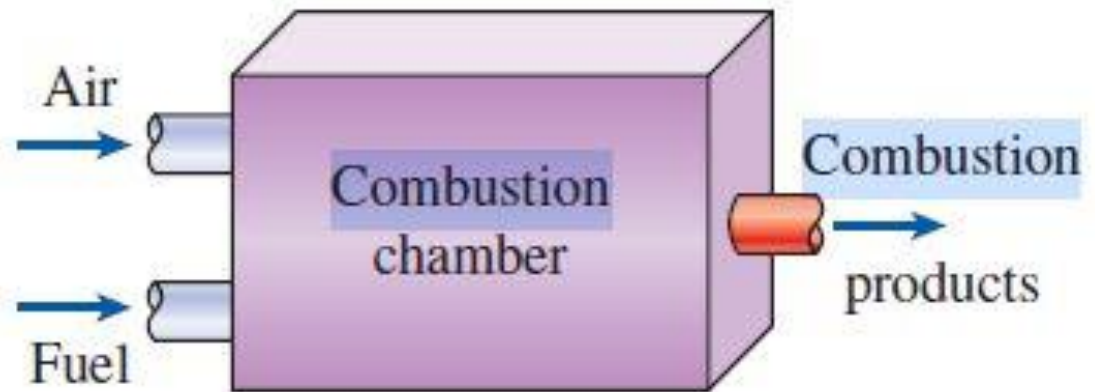
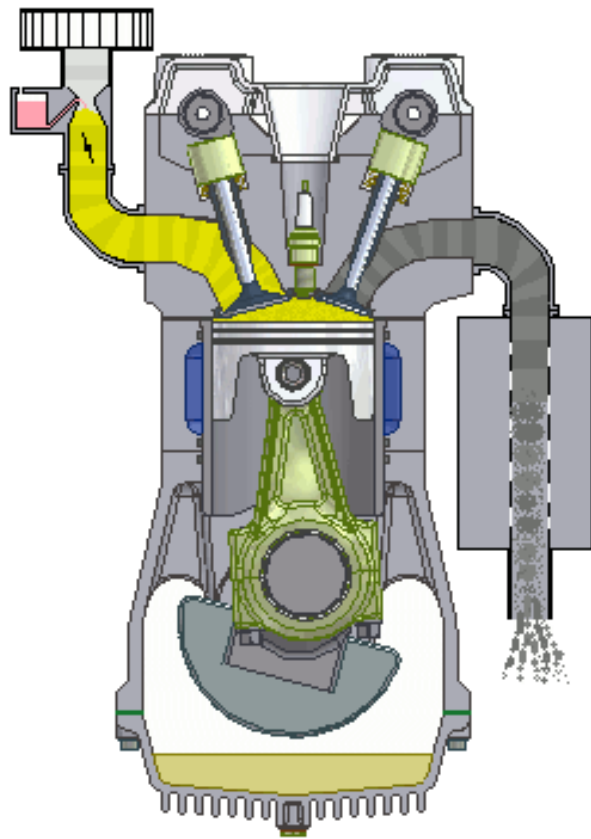


(a) Fission of uranium



(b) Fusion of hydrogen

Chemical Energy

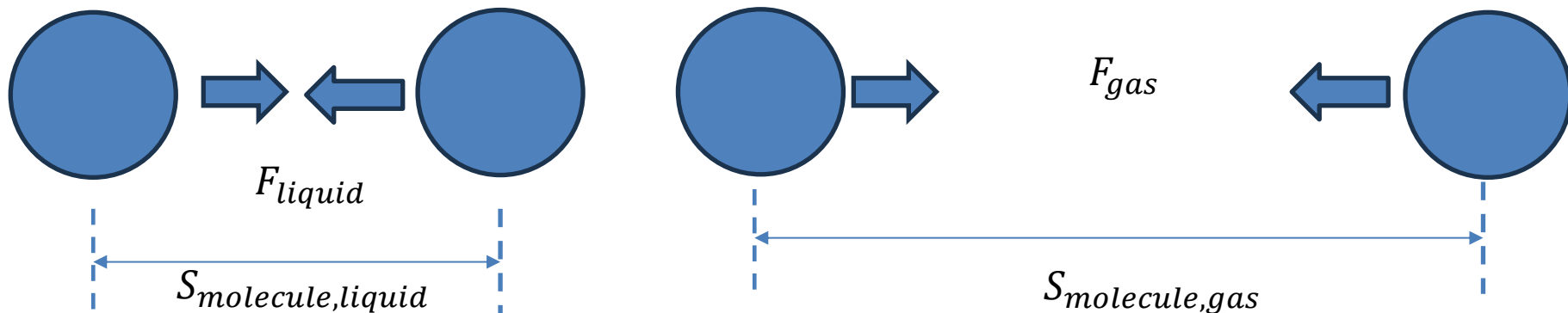


	$\text{CH}_4 + 2(\text{O}_2 + 3.76\text{N}_2) \rightarrow$
	$\text{CO}_2 + 2\text{H}_2\text{O} + 7.52\text{N}_2$

Intermolecular Potential Energy (For an ideal gas = 0)

The internal energy is also associated with various *binding forces* between the molecules of a substance, between the atoms within a molecule, and between the particles within an atom and its nucleus. The forces that bind the *molecules* to each other are, as one would expect, strongest in solids and weakest in gases. If sufficient energy is added to the molecules of a solid or liquid, the molecules overcome these molecular forces and break away, turning the substance into a gas. This is a phase-change process. Because of this added energy, a system in the gas phase is at a higher internal energy level than it is in the solid or the liquid phase. The internal energy associated with the phase of a system is called the *latent energy*. The phase-change process can occur without a change in the chemical composition of a system. Most practical problems fall into this category, and one does not need to pay any attention to the forces binding the atoms in a molecule to each other.

$$F_{gas} < F_{liquid}; S_{gas} > S_{liquid}; U_{potential,gas} > U_{potential,liquid}$$

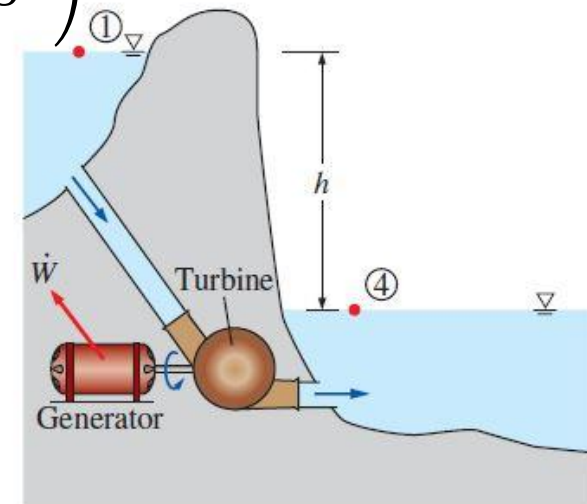


The 1st law of thermodynamics talks about the **quantitative conservation of total energy** which includes the, Thermal, Mechanical, Chemical, Nuclear, together.

The **Bernoulli's equation** talks about the conservation of **Mechanical energy** which includes the **Bulk kinetic**, **Bulk Potential** and the Pressure (**flow energy**). Out of these three forms, the **Bulk Kinetic and Bulk Potential energies** are stored form energy. However, the fluid can transfer pressure energy but can not store it. A Pump transfer mechanical energy to a fluid to increase its pressure and Turbine extracts mechanical energy from the fluid by dropping its pressure.

$$\text{Mechanical Energy } \dot{E}_{mech} = \dot{m}e_{mech} = \dot{m} \left(Pv + \frac{1}{2}V^2 + gz \right)$$

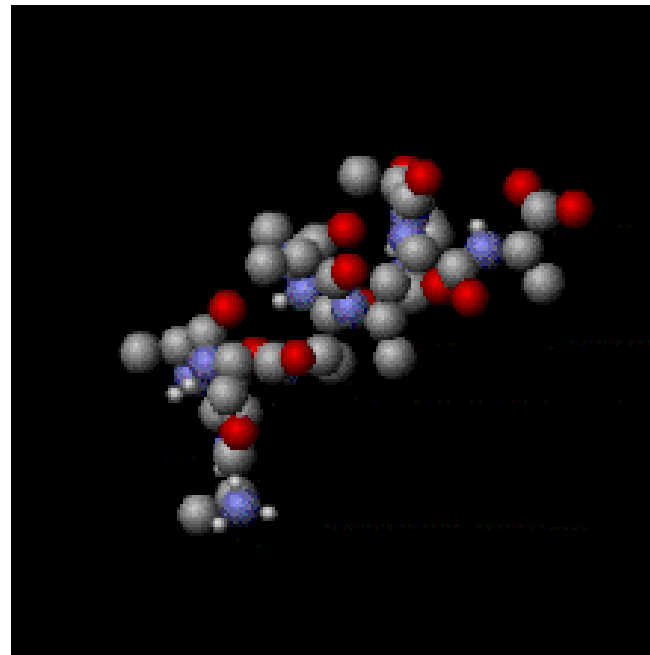
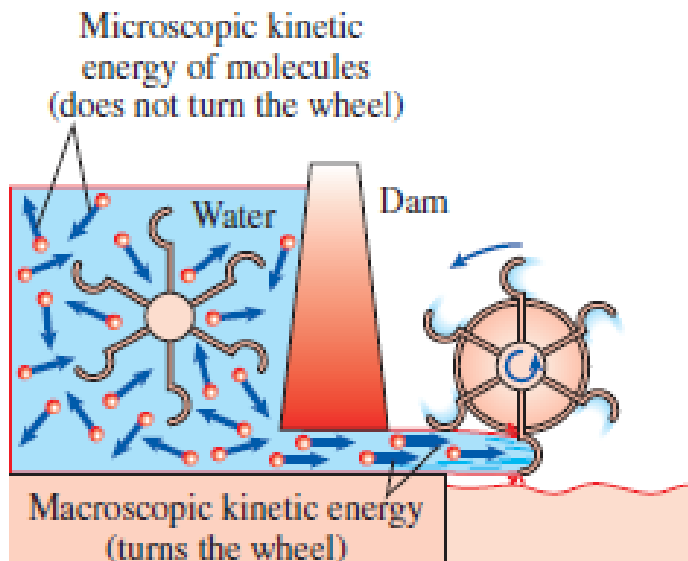
High grade Energy (Ordered Form of Energy)



The **mechanical energy** can be defined as the form of energy that **can be converted to mechanical work completely** and **directly** by an ideal mechanical device such as an ideal turbine.

Thermal energy is not a mechanical energy as it can not be converted directly and completely in continuous fashion to mechanical work (this is the constraint imposed by 2nd law of Thermodynamics)

Ordered & Disordered Form of Energy



End of Module 1